Continuous-Time Reinforcement Learning for Portfolio Optimization under Partial Information

A Review of Wu and Li (2024)

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The Pitch

Outline

1. Introduction & Motivation
2. Background
3. Methods: The RSEMV Problem & POEMV
4. Simulation Study
5. Conclusion & Open Problems

Quick Roadmap

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Exploratory Framework

Wang, Zariphopoulou, et al. (2020)

→

2

Application EMV

Wang and Zhou (2020)

→

3

Application POEMV

Wu and Li (2024)

Motivation

Why continuous-time RL?

- Financial markets evolve continuously, so discrete-time RL has poor motivation.
- Key challenge of continuous time: explore without collapsing to a deterministic policy.

Why regime switching?

- Markets exhibit bull/bear non-stationarity, which is qualitatively important.
- EMV ignores regimes.

Figure: S&P 500 (1992–2022): bull (green) / bear (red).

RL vs. the Classical Approach

Classical (Plug-In) Approach

1. Estimate model parameters (mean returns) from historical data.
2. Substitute estimates directly into the optimal control formula.

Issue: Mean return estimates are noisy, and small errors can produce large swings in holdings, with no built-in self-correcting mechanism.

RL Approach

1. Learn and update policy directly through market interactions.

Advantage: Policy self-corrects; estimation uncertainty is contained within the (belief) state rather than propagating directly into portfolio decisions.

Wu & Li (2024): POEMV

Wang and Zhou (2020) established EMV for continuous-time MV optimization; Wu and Li (2024) extend this to partial information and regime switching via **POEMV**.

The MV Portfolio Problem

Over a fixed horizon $[0, T]$, we want to minimize terminal wealth variance subject to a mean target:

$$\min_u \ \mathrm{Var}[X_T^u] \quad \text{s.t.} \quad \mathbb{E}[X_T^u] = z, \quad \text{for admissible } u.$$

Via a Lagrange multiplier ω, this is equivalent to:

$$\min_u \ \mathbb{E}\big[(X_T^u - \omega)^2\big] - (\omega - z)^2.$$

- **Time-inconsistent**: the variance term prevents a dynamically optimal solution.
- Wu and Li (2024) focus on the **precommitted strategy**, optimal only at $t = 0$.
- The multiplier ω^* is found via a **stochastic approximation**, with updates every $s = 10$ iterations to avoid destabilizing the policy learning loop.

Exploratory Stochastic Control

Key idea: replace point-action controls u_t with *distribution-valued* policies $\boldsymbol{\pi}_t \in \mathcal{P}(\mathcal{U})$.

The **wealth process** under stochastic policy $\boldsymbol{\pi}$:

$$d\tilde{X}_t^{\boldsymbol{\pi}} = \underbrace{\sigma \hat{\rho}_t \, \mathbb{E}_{\boldsymbol{\pi}}[u]}_{\text{mean drift}} \, dt + \underbrace{\sigma \, \mathbb{E}_{\boldsymbol{\pi}}[u]}_{\text{env. noise}} \, d\hat{B}_t + \underbrace{\sigma \sqrt{\mathrm{Var}_{\boldsymbol{\pi}}[u]}}_{\text{action noise}} \, d\bar{B}_t$$

where $\hat{B}, \bar{B}$ are independent Brownian motions.

Entropy-regularized objective (temperature $\Lambda \geq 0$):

$$\min_{\boldsymbol{\pi}} \; \mathbb{E}\left[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda \int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot \mid t, X_t, p_t)) \, dt \right] - (\omega - z)^2$$

- $\Lambda \to 0$ recovers deterministic optimal control.
- For LQ problems: optimal $\boldsymbol{\pi}^*$ is **Gaussian** with tractable mean and variance.
- Extending to regime switching couples the wealth and belief dynamics, requiring additional structure.

Market Model & Wonham Filtering

Regime-switching asset price:

$$dS_t = S_t(\mu_t\,dt + \sigma\,dB_t), \quad \mu_t \in \{\mu_1, \mu_2\}$$

Regime $I_t \in \{1, 2\}$ follows a continuous-time Markov chain with rates λ_1, λ_2.

Self-financing wealth process (dollars invested u_t, risk-free rate r):

$$dX_t = rX_t\,dt + u_t\big[(\mu_t - r)\,dt + \sigma\,dB_t\big] = rX_t\,dt + u_t\sigma\big[\rho_t\,dt + dB_t\big]$$

Belief state: $p_t = P(I_t = 1 \mid \mathcal{F}_t^S)$, with:

$$dp_t = \big[-(\lambda_1{+}\lambda_2)p_t + \lambda_2\big]dt + (\rho_1{-}\rho_2)p_t(1{-}p_t)\,d\hat{B}_t$$

where the **per-regime Sharpe ratios** $\rho_i = (\mu_i - r)/\sigma$ are fixed constants, and
the **filtered Sharpe ratio** $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$ is the belief-weighted expectation $\mathbb{E}[\rho_t \mid \mathcal{F}_t^S]$.

Separation Principle

Key idea: p_t is a *sufficient statistic* for μ_t, so the partial-information problem reduces to a fully observed control problem in (X_t, p_t).

- Substituting $\hat{\rho}_t$ for ρ_t, no hidden state remains:

$$dX_t = \sigma u_t(\hat{\rho}_t \, dt + d\hat{B}_t)$$

- This yields a tractable, fully observed HJB equation in (X_t, p_t).
- $p_t \in (0, 1)$ strictly, so no boundary conditions are needed and standard martingale arguments apply.

Value Function Structure (Theorem 1)

The value function for the RSEMV problem takes the form

$$v(t,x,p) = \frac{(x-\omega)^2}{f(t,p)} + g(t,p) - (\omega - z)^2,$$

decoupling the problem into two scalar functions of (t,p) alone.

Key structural facts

- f and g satisfy **degenerate linear parabolic PDEs** in (t,p), solved numerically; degeneracy at $p \in \{0,1\}$ is harmless (entrance boundaries).
- f encodes **exploration variance**; g encodes the **entropy bonus**.
- f is bounded and decreasing in t; well-posedness follows.

PDE details in Appendix.

Optimal Policy: Myopic–Hedging Decomposition (Theorem 2)

Given the value function structure of Theorem 1, minimizing over $\boldsymbol{\pi}$ yields a **Gaussian optimal policy**:

$$\boldsymbol{\pi}^*(\cdot \mid t, x, p) = \mathcal{N}\left(m^*(t,x,p),\ \frac{f(t,p)\Lambda}{2\sigma^2}\right)$$

where the **mean decomposes** as:

$$m^*(t,x,p) = \underbrace{-\frac{x-\omega}{\sigma}\hat{\rho}_t}_{\text{myopic: current filtered Sharpe ratio}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1-\rho_2)p(1-p)\frac{f_p}{f}}_{\text{hedging: regime transition risk}}$$

- Hedging demand is **largest at** $p_t \approx \frac{1}{2}$ (maximum uncertainty) and vanishes when beliefs concentrate near 0 or 1.
- Setting $\rho_1 = \rho_2$ eliminates hedging and recovers the single-regime case.
- Policy variance $\propto \Lambda f(t,p)$ decays as $t \to T$: exploration naturally winds down near the investment horizon.

The POEMV Algorithm

Parametrize $\ln f^\theta$ and g^θ as order-m polynomials in p and $(T-t)$. The resulting policy is Gaussian with mean and variance determined by (θ, ϕ):

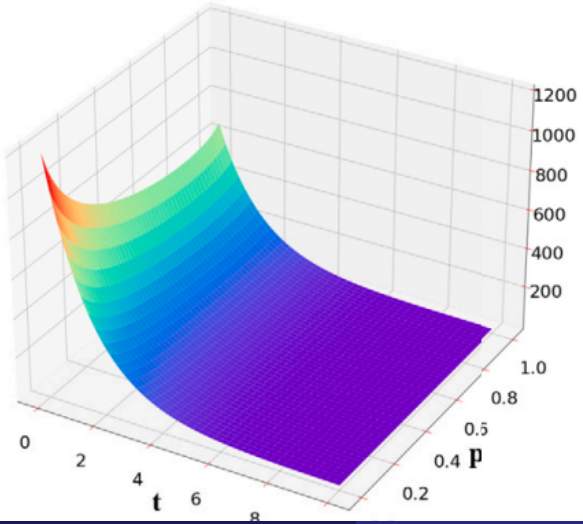
$$\boldsymbol{\pi}^\phi(\cdot \mid t, x, p) = \mathcal{N}\left(-(x-\omega)\left[\phi_1\left(p - \frac{f_p^\theta}{f^\theta} p(1-p)\right) + \phi_2\right],\ \Lambda e^{\phi_3} f^\theta(t,p) \right)$$

Each episode (one full simulated path over $[0, T]$):

1. **Filter:** compute p_{t_k} via discretized Wonham filter.
2. **Critic (PE):** gradient descent on θ minimizing the martingale loss.
3. **Actor (PG):** score-function gradient step on ϕ.

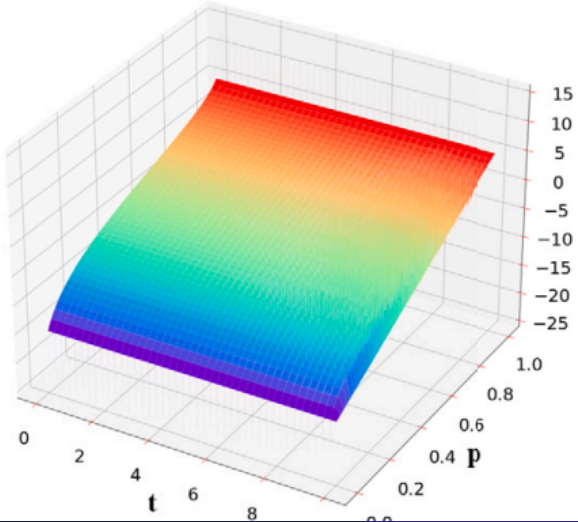
Every 10 episodes — Lagrange update: $\omega \leftarrow \omega - \eta_\omega(\bar{X}_T - z)$, enforcing the mean-wealth constraint on a slower timescale.

PDE Surface: $f(t,p)$



- $f(t,p)$ appears in the value function as $V^* = \frac{(x-\omega)^2}{f(t,p)} + g(t,p) - (\omega - z)^2$.
- Decreasing as a function of t, so exploration shrinks as the investment end date approaches.

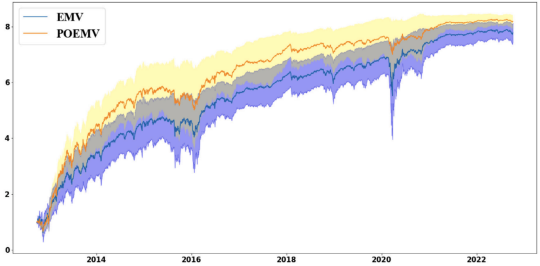
PDE Surface: $\gamma^*(t,p)$



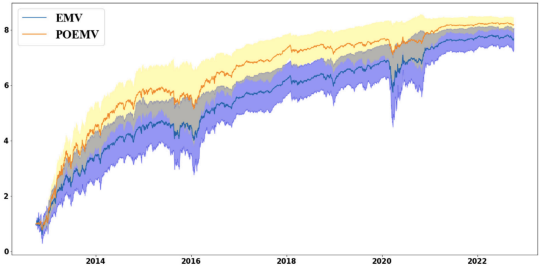
- $\gamma^*(t,p)$ is the **wealth-scaled investment coefficient**: $m^*(t,x,p) = -(x-\omega)\,\gamma^*(t,p)$.
- Increases as a function of p, so invest more in the risky asset when probability of bull-market belief increases.

Wu & Li (2024): S&P 500 Empirical Results

30-year S&P 500 data; 20-year train / 10-year test. 30 $\beta \approx 1$ stocks as index proxies (≈75k episodes).



(a) Daily rebalancing



(b) Monthly rebalancing

Our Simulation Study

DAX Application

The DAX is Germany's primary stock index (30 blue-chip firms); we apply and compare POEMV and EMV in this setting.

	DAX	S&P 500
μ_1	0.029	0.250
μ_2	-0.081	-0.730
σ	0.115	0.220
λ_1	0.110	0.360
λ_2	0.339	2.890
r	0.015	0.000

Table: DAX vs. S&P 500 calibrated parameters

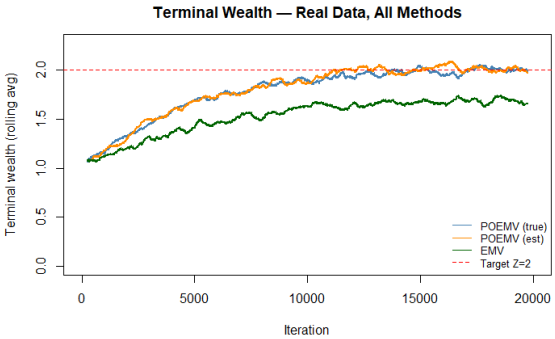
Key differences:

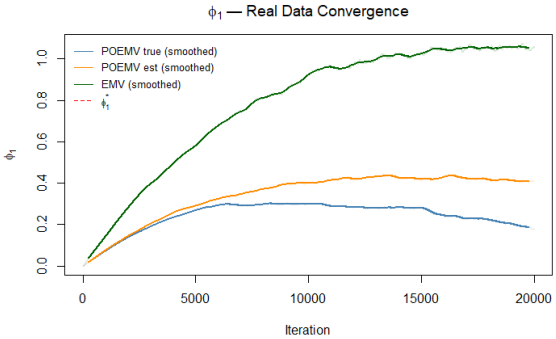
- Smaller drifts and bear regimes lasting $\sim$3 years (vs. $\sim$4 months).
- Non-zero risk-free rate ($r = 0.015$) from ECB deposits.
- Only 8 training paths survive the β-filter.

DAX Training Dynamics

Potential Issue

Because we only have 8 training paths, we likely haven't achieved convergence.





DAX Out-of-Sample Results

Initial Results

Results are overall quite surprising compared to Wu and Li; motivates further exploration.

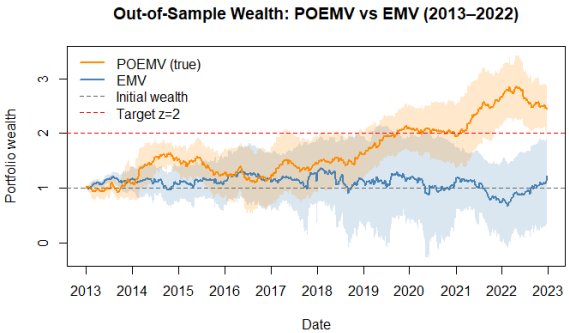


Table: OOS metrics (DAX)

	Mean	Var	Sharpe
POEMV (true)	1.070	1.171	0.06
POEMV (est.)	1.460	0.899	0.49
EMV	1.043	0.787	0.05
Target z	2.000	—	—

Conclusion & Open Problems

Takeaways

- Wonham filtering + separation principle make the martingale actor–critic viable under partial information.
- Regime-awareness materially matters: POEMV halves out-of-sample variance vs. EMV.
- DAX results are encouraging but training data scarcity is a real constraint.

Open Problems

- **Joint estimation & control:** filtering error dominates as the binding constraint.
- **Scalability:** multiple assets and regimes require revisiting both filter and HJB.

References I

- Wang, Haoran, Thaleia Zariphopoulou, and Xun Yu Zhou (2020). "Reinforcement Learning in Continuous Time and Space: A Stochastic Control Approach". In: *Journal of Machine Learning Research* 21.198, pp. 1–34.
- Wang, Haoran and Xun Yu Zhou (2020). "Continuous-Time Mean–Variance Portfolio Selection: A Reinforcement Learning Framework". In: *Mathematical Finance* 30.4, pp. 1273–1308. DOI: 10.1111/mafi.12281.
- Wu, Bo and Lingfei Li (2024). "Reinforcement Learning for Continuous-Time Mean-Variance Portfolio Selection in a Regime-Switching Market". In: *Journal of Economic Dynamics and Control* 158, p. 104787. DOI: 10.1016/j.jedc.2023.104787.

Appendix: Discretized Wonham Filter

Apply Itô's formula and Euler scheme to $q_t^1 = \log p_t$ and $q_t^2 = \log(1 - p_t)$ to guarantee $p_{t_k} \in (0, 1)$:

$$q_{t_k}^1 = q_{t_{k-1}}^1 + \left[\tilde{\lambda}_2 e^{-q_{t_{k-1}}^1} - (\tilde{\lambda}_1 + \tilde{\lambda}_2) - \tfrac{1}{2}(\tilde{\rho}_1 - \tilde{\rho}_2)^2 (1 - e^{q_{t_{k-1}}^1})^2\right] \Delta t + (\tilde{\rho}_1 - \tilde{\rho}_2)(1 - e^{q_{t_{k-1}}^1}) \Delta \hat{B}_{t_k}$$

Then: $p_{t_k} = e^{q_{t_k}^1} / (e^{q_{t_k}^1} + e^{q_{t_k}^2})$, guaranteed in $(0, 1)$.

Innovation process increment:

$$\Delta \hat{B}_{t_k} = \frac{1}{\tilde{\sigma}} \left[\log \frac{S_{t_k}}{S_{t_{k-1}}} - (\hat{\mu}_{t_{k-1}} - \tilde{\sigma}^2/2)\Delta t\right], \quad \hat{\mu}_{t_{k-1}} = \tilde{\mu}_1 p_{t_{k-1}} + \tilde{\mu}_2 (1 - p_{t_{k-1}})$$

This log-transform discretization is used in both the Wu & Li S&P 500 implementation and our DAX application, guaranteeing valid probabilities at every time step regardless of step size.

Appendix: POEMV Gradient Updates

PE update ($\Delta\theta$, martingale loss gradient):

$$\Delta\theta = \sum_{k=0}^{K-1}\left[\left(h(X_T) - V^\theta(t_k, X_{t_k}, p_{t_k}) - \sum_{i=k}^{K-1}\Lambda\mathcal{H}(\boldsymbol{\pi}_{t_i}^\phi)\Delta t\right)\frac{\partial V^\theta}{\partial\theta}(t_k, X_{t_k}, p_{t_k})\right]\Delta t$$

PG update ($\Delta\phi$, score-function gradient):

$$\Delta\phi = \sum_{k=0}^{K-1}\frac{\partial}{\partial\phi}\ln\boldsymbol{\pi}^\phi(u_{t_k} \mid t_k, X_{t_k}, p_{t_k})\left[V_{t_{k+1}}^\theta - V_{t_k}^\theta - \Lambda\mathcal{H}(\boldsymbol{\pi}_{t_k}^\phi)\Delta t\right] - \Lambda\frac{\partial\mathcal{H}}{\partial\phi}\Delta t$$

Appendix: Estimation Method Comparison (Table 2 of Wu and Li (2024))

Table: Heuristic vs. EM algorithm (100 simulated paths)

	μ_1	μ_2	λ_1	λ_2	σ
True	0.25	-0.73	0.36	2.89	0.22
Heuristic	0.58 (0.10)	-0.69 (0.13)	1.21 (0.42)	2.09 (0.59)	0.22 (0.003)
EM	0.51 (0.65)	-0.21 (0.28)	128.59 (90.14)	132.21 (88.03)	0.25 (0.12)

- EM fails on transition rates.
- Heuristic adopted throughout. The same failure mode applies to DAX data, motivating the heuristic approach there too.
- Both methods estimate σ accurately.